

Objectives

- Principles and Techniques used in Musculoskeletal Imaging
- Vocabulary for Describing Bony Anatomy
- Fracture and dislocation Language, how to communicate information about a fracture
- Language to Describe Bone Lesions
- Recognize various osseous pathology on radiographs
- Periosteal changes, appearance and their significance

Basic approach to bone films

- Systematic approach
 - Most important- Identifying the body part 😊
 - Identifying the views
 - The standard views will be at **minimum** frontal and lateral views
 - Most joints will also include an oblique view/oblique variant
 - Always evaluate if the patient is an adult or a child as your radiographic differential diagnosis will change based on the age of the patient

Basic approach to bone films

- Step 1
 - Checking alignment of the osseous structures
- Step 2
 - Evaluate joint spacing
 - Do they appear narrowed or widened? How does the joint in question compare with other joints visible on the film?
- Step 3
 - Evaluate the bone cortex it should be smooth and continuous
 - Nutrient foramina- long bone via an obliquely oriented canal. They are often observed with a well-corticated edge
 - Physeal plates and sesamoids
- Step 4
 - Evaluate the surrounding soft tissue structures!!
 - Is there soft tissue swelling? Joint effusion?

Maximum x-ray
Transmission
(least dense
tissue)

Darkest

air

fat

soft tissue/fluid*

calcium

bone

x-ray contrast

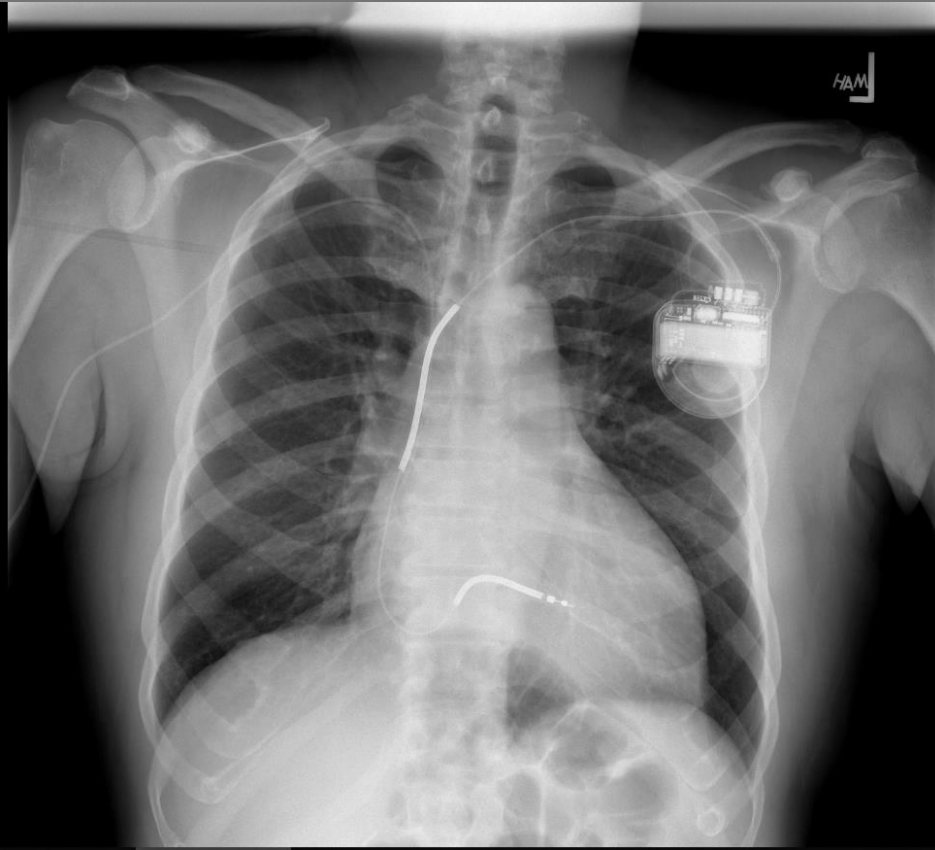
metal

Maximum x-
ray

Whitest

Absorption
(densest tissue)

*fluid can be pus, blood or
simple fluid



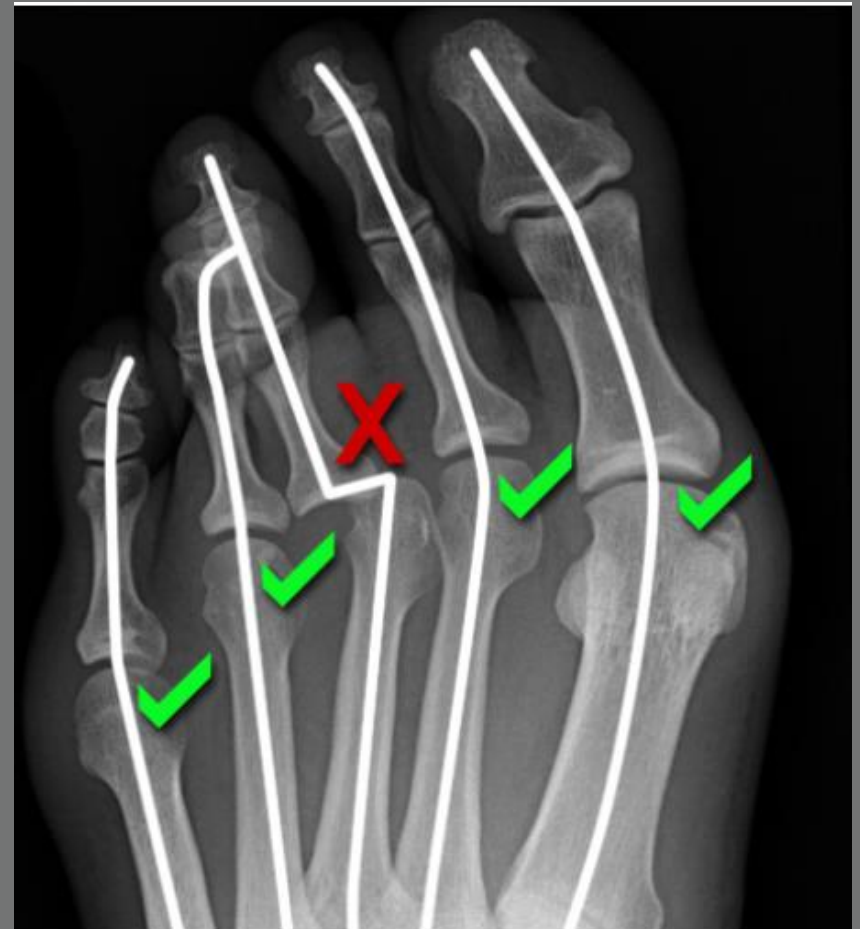
Joint alignment

- Loss of alignment may be due to a fracture or a joint dislocation.



Joint alignment

- Loss of alignment may be due to a fracture or a joint dislocation.



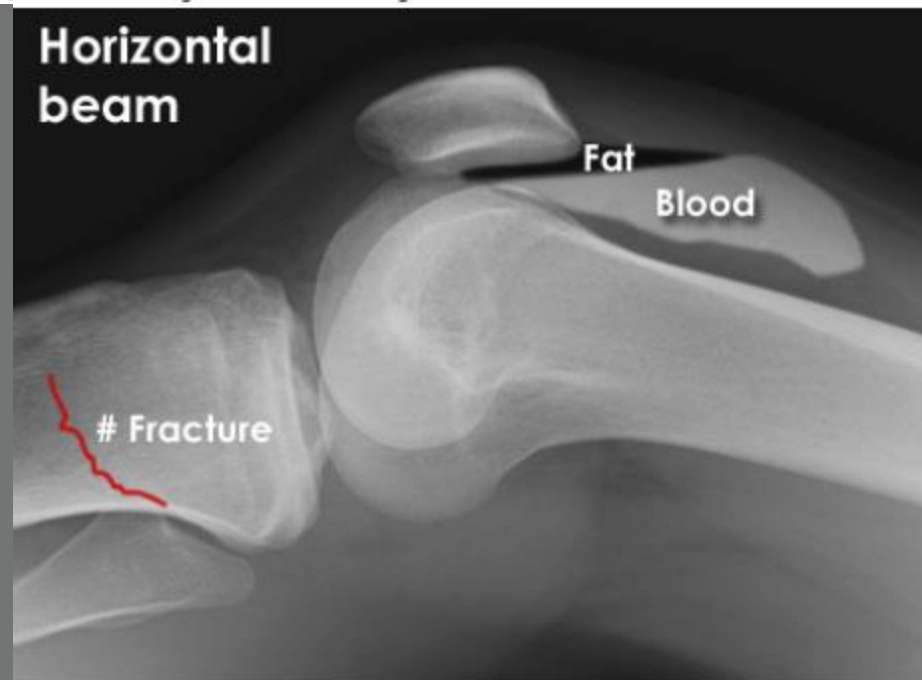
Soft tissues *

- Not uncommonly an abnormality of soft tissues is more obvious than a bone injury, or may even imply a bone injury that is not visible at all.
- Look for focal soft tissue swelling as well as the “joint spaces” themselves for effusions



Soft tissues

- Not uncommonly an abnormality of soft tissues is more obvious than a bone injury, or may even imply a bone injury that is not visible at all



Viewing principles

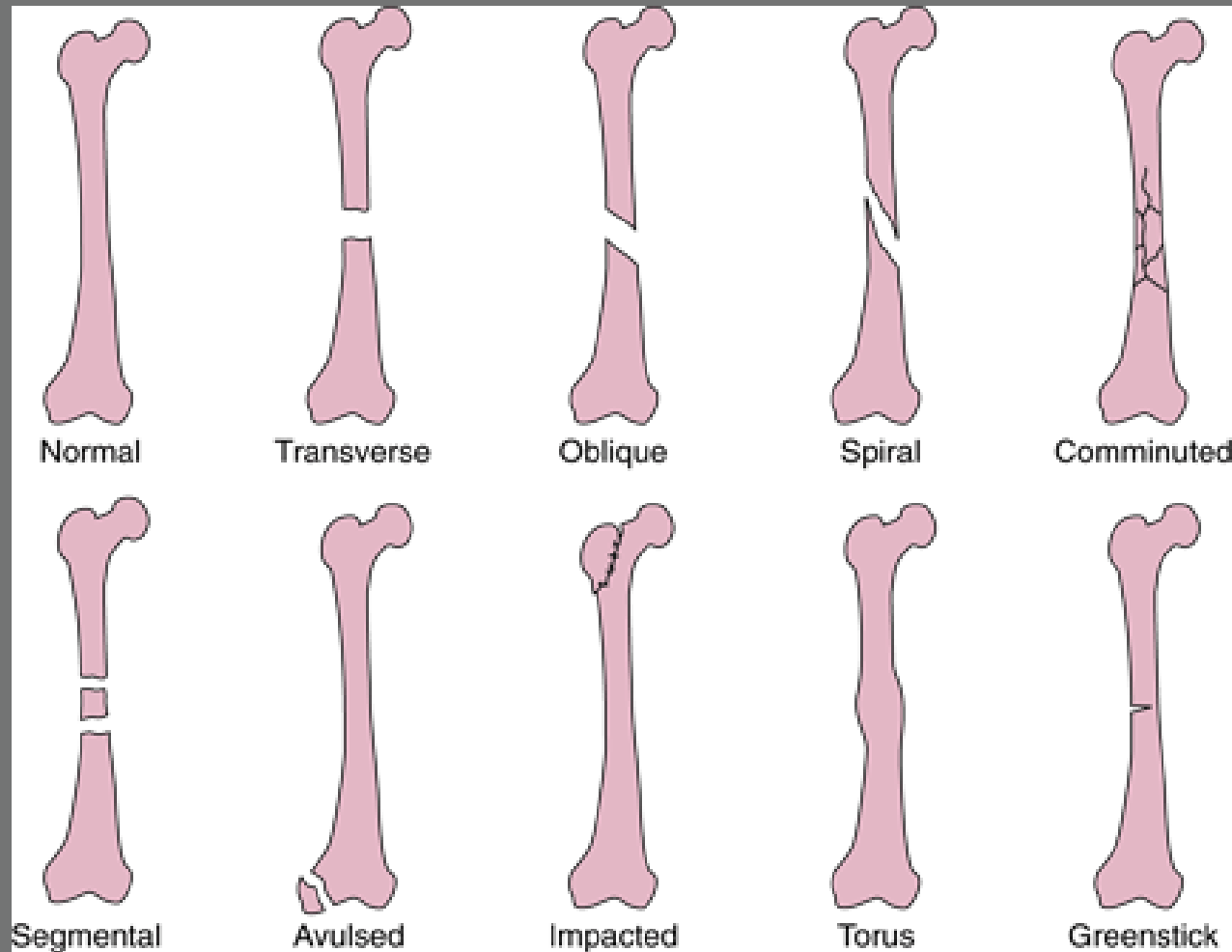


Fracture Description

- **Open vs Closed:** fracture is open when exposed to air (laceration or gross exposure)
- **Pathologic fracture:**
 - Normal stress on abnormal bone
 - Implies fracture through weakened bone i.e a bone lesion, severe osteopenia
- **Stress fracture:**
 - Implies misuse or overuse. Abnormal stress on normal bone

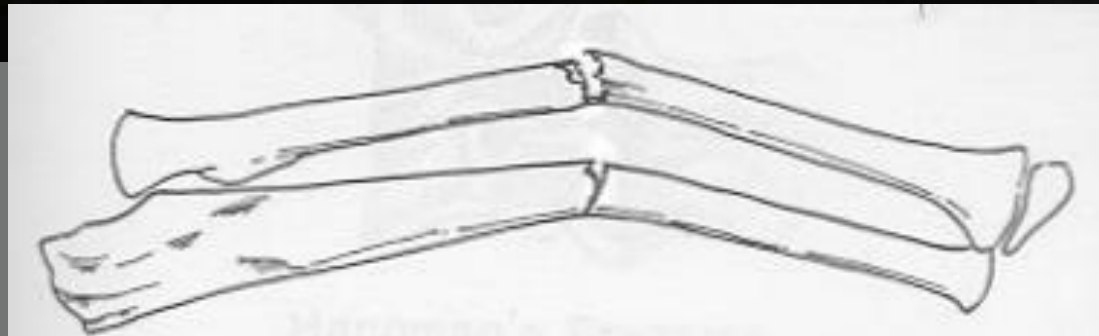


Path of the Fracture



Greenstick fracture

- Cortex broken on one side of the bone and only bent or buckled on the other side



Stress Fracture



Stress Fracture





- Benign and non-growing (or extremely slowly growing) lesions are well circumscribed and show **geographic pattern** of bone destruction with a **sclerotic rim**.
 - Geographic pattern refers to a **well-defined** area of lysis.
 - Well defined in that I can take a pencil and trace around the lesion.
 - The sclerotic rim is more commonly seen in the weight-bearing bones and represents bone reaction to the lesion.
 - Its presence means that the bone has been given sufficient time to react.
 - Some authors say that the sclerotic rim signifies benignancy to about 95%.

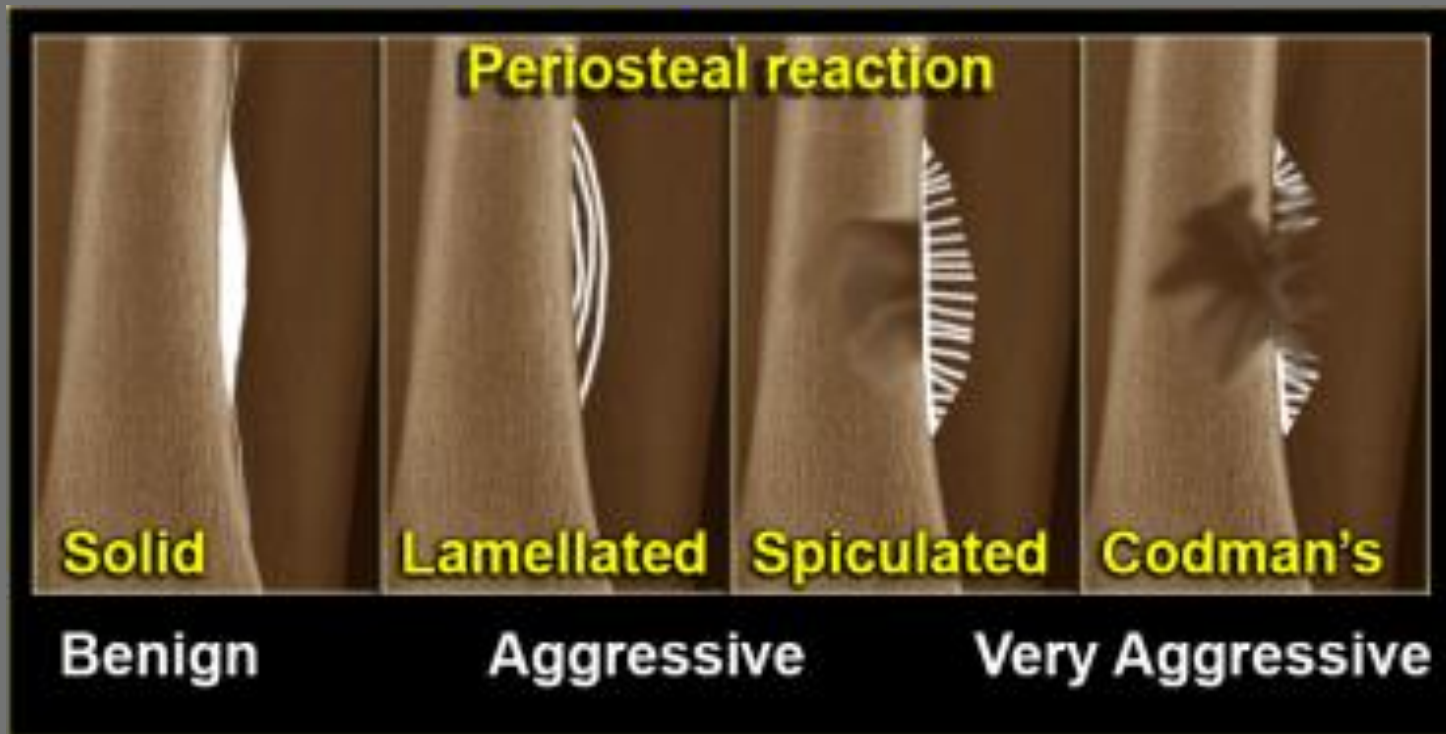
- If the lesion is growing more rapidly, it may still show a well-demarcated zone of bone destruction (geographic pattern), but it will lack a sclerotic rim.
- With continued growth, such lesions may show **cortical expansion**.
- Expansile growth pattern is defined as visible widening of the affected portion of bone.
- This pattern may be seen in locally aggressive tumors and in malignancies.





- Rapidly growing lesions are poorly defined and may show aggressive, infiltrative patterns of bone destruction (**permeative or "moth-eaten"**).
- **"Moth-eaten" pattern** is defined as an ill-defined zone of multiple small radiolucencies that may coalesce.
- They are seen in **high-grade malignant neoplasms and in osteomyelitis**

Periosteal Reactions

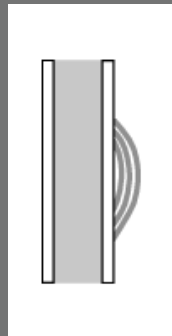


Periosteal reaction may be classified as benign or aggressive (note: not benign and malignant) based on the time course of the initiating process.

Periosteal Reactions



- Rapidly growing processes, the periosteum **cannot** produce new bone as fast as the lesion is growing.
- Therefore, rather than a solid pattern of new bone formation, we see an interrupted pattern.
- This interrupted pattern can manifest itself in several ways, depending on just how steadily the lesion grows.
- If the lesion grows sporadically, then the periosteum may have time to lay down a thin shell of calcified new bone before the lesion takes off again on its next growth spurt.
- This may result in a pattern of one or more concentric shells of new bone over the lesion.
- This pattern is sometimes called lamellated or “onion-skin” periosteal reaction.



Periosteal Reactions

- If the lesion grows rapidly but steadily, the periosteum will not have enough time to lay down even a thin shell of bone, and the pattern may appear quite different.
- In such cases, the tiny fibers that connect the periosteum to the bone (Sharpey's fibers) become stretched out perpendicular to the bone.
- When these fibers ossify, they produce a pattern sometimes called “sunburst” or “hair-on-end” periosteal reaction, depending of how much of the bone is involved by the process.

